

StormGuard

Flood situational awareness for emergency responders

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Code, data sources and algorithms: <https://github.com/mcmanoj2001/StormGuard>

Algorithm descriptions: <https://github.com/mcmanoj2001/StormGuard/blob/main/docs/ALGORITHMS.md>

Summary

StormGuard is a browser-based flood situational-awareness tool for emergency responders. It combines live USGS river gauges, NWS alerts and river-crest forecasts, NHC hurricane tracks, NEXRAD radar, FEMA flood zones, Census population and OpenStreetMap hospitals on one map, and ranks what needs attention as **Respond now / Prepare / Monitor**. Each warning shows the people living in the area, how many are age 65+, homes without a vehicle, and hospitals inside it. Every data source is cached and fails independently, so one outage never blanks the others, and the last good data is shown flagged “Data delayed”. A built-in test scenario injects data for demonstrations when no disaster is active. It runs entirely in the browser with no login, backend or database, and adapts to desktop, tablet and phone widths.

Key capabilities

- Live data with a visible refresh time and per-gauge reading times; the status badge updates as the app refreshes every 60 seconds.
- Hovering and clicking gauges and warnings; clicking a map event filters the action list to that event.
- Ranked actions with per-warning impact: people in the area, age 65+, homes without a vehicle, hospitals.
- Marking actions done, a built-in test scenario, an in-app list of every data source and its limits, and a responsive layout.

Data sources and how to access them

Source	Endpoint (public, no login)	Refresh	Used for
USGS stream gauges	waterservices.usgs.gov/nwis/iv/	5 min	Stage, rate of rise/fall, severity
NWS alerts + zones	api.weather.gov/alerts/active ; api.weather.gov/zones	60 s	Warnings and their map areas
NWS river forecasts	api.water.noaa.gov/nwps/v1/gauges/{id}	6 h	Crest forecast, official flood category (13 points)
NHC hurricane	mapservices.weather.noaa.gov/tropical/rest/services/tropical/NHC_tropical_weather/MapServer	10 min	Track, cone, early-warning card
NEXRAD radar	mesonet.agron.iastate.edu/cache/tile.py/1.0.0/nexrad-n0q-900913/	live	Precipitation overlay
FEMA flood zones	hazards.fema.gov/arcgis/rest/services/public/NFHL/MapServer (layer 28)	tiles	Flood-zone overlay
Census population	api.census.gov/data/2022/acs/acs5 (free key)	24 h	People, age 65+, no-vehicle homes, mobile homes
Census boundaries	tigerweb.geo.census.gov/arcgis/rest/services/TIGERweb	24 h	Tract and state geometry
OpenStreetMap hospitals	overpass-api.de/api/interpreter	24 h	Hospitals inside warnings

All sources are free federal or public services called directly from the browser. The Census population request needs a free key (api.census.gov/data/key_signup.html); everything else needs none.

How derived data is generated

- **Rate of rise/fall:** latest stage minus the reading about 1 hour earlier, in ft/hr. Rising above +0.1, falling below -0.1.
- **Gauge severity:** fixed stage thresholds (30 / 20 / 12 ft for major / moderate / minor). Not each gauge's official flood stage; the 13 NWS forecast points use NWS's own categories.
- **Alert areas:** most NWS alerts carry only zone codes, so each zone's boundary is fetched and combined into one polygon per alert.
- **Population estimate:** a census tract counts when its center lies inside the warning polygon (point-in-polygon); its whole population is counted. Totals are de-duplicated across overlapping warnings. This is people living in warned zones, an upper bound on exposure.
- **Hospitals at risk:** OpenStreetMap hospitals inside a severe or extreme flood-related warning polygon.
- **Respond now / Prepare / Monitor:** rule-based tiers from warning severity, gauge level and trend, forecast crest timing, and the population and hospital counts inside the area. Only flood-related warnings count.

Full descriptions with the exact rules: <https://github.com/mcmanoj2001/StormGuard/blob/main/docs/ALGORITHMS.md>

Test-data injection

A "Test scenario" toggle replaces live responses with a sample flood-and-hurricane event inside the same fetch, normalize, rank and render pipeline the live data uses, so demonstrations exercise the real algorithms. The extra demographic figures in that sample are illustrative values, not Census data.

Known limitations

- No evacuation orders: no federal feed carries them. "Respond now" means the highest-priority warning, not an evacuation order.
- Population figures count whole census tracts, so they are upper-bound estimates.
- Gauge severity uses fixed thresholds, so some high-datum gauges (for example reservoirs) can read "major".
- Radar only: no satellite imagery. No real-time power or cell outage data (no federal API exists).
- LiDAR river-stage sensors exist, but no public national feed does.

Run it

Clone <https://github.com/mcmanoj2001/StormGuard>, then run `npm install` and `npm run dev` and open <http://localhost:5173>. The first live load can take up to a minute because the USGS service is slow; afterwards data is cached.